

BPM Report

Group #4

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This report highlights our group's collective efforts in the field of business process management, showcasing a comprehensive approach to identifying and improving organizational processes through the strategic application of information technology (IT). For this project, we focused on the maintenance departments at Miami University as the subject of our study. Our group focused on this process because various group members have experienced or heard about unfavorable interactions with maintenance. We then worked collaboratively to develop two solutions to minimize these interactions by increasing the simplicity and transparency for all parties involved.

Our work began with an in-depth examination of an existing business process within the department, interviewing a specialist, and identifying areas where IT integration could drive meaningful improvements. We then meticulously modeled the current process, explored a range of potential enhancements, and selected the most effective IT-based solution for

implementation. Following this, we developed a detailed model of the improved process and performed a thorough cost-benefit analysis to evaluate the feasibility and anticipated impact of our proposed solution.

Problem Description

As students at Miami University, living and studying on campus is a unique and rewarding experience. However, it is not without its occasional challenges, such as dealing with broken dressers, malfunctioning appliances, or leaky shower heads. Fortunately, the university provides a maintenance request system to help address these issues and simplify our daily lives. While this system is designed to make it easier for students to resolve such problems, we have observed that the process often falls short in terms of organization and communication, which can lead to unnecessary confusion and frustration.

Currently, when a student submits a maintenance request, they receive an automatic confirmation email acknowledging the submission. This initial step provides reassurance that the request has been logged. The maintenance portal then updates to display a status page indicating that the request is “pending.” At this point, however, students are left in the dark regarding when their issue will be addressed, with no clear timeline or updates to keep them informed. Eventually, the student receives another email with the scheduled date and a broad time frame for when the maintenance worker is expected to arrive, along with instructions on how to prepare for the visit. This email also mentions that the maintenance worker will call upon arrival. In

some cases, the maintenance team has left this step out, creating confusion for students.

While this process provides a basic framework, it lacks transparency and real-time communication, leaving students uncertain about the progress of their requests. The ambiguity surrounding timelines and the limited proactive updates can result in unnecessary stress, especially for students juggling busy schedules. Addressing these shortcomings presents an opportunity to significantly improve the maintenance experience for Miami University students, fostering a more efficient and student-friendly process.

On the other side of the equation, the maintenance process is cluttered and confusing. In order to understand this process, we interviewed Daniel Olthaus, the Director of Maintenance at Miami University. Olthaus shared every step of the maintenance process, walking us through the eight swim lanes that include separate teams for planning, scheduling, and operations in addition to the expected roles of student, technician, and management.¹

Click the [link](#) to the Lucid Chart Business Process Model and Notation to see the current process flow for a maintenance request.

Solution Design

In designing our solutions, we wanted to focus on eliminating confusion by improving organization, process complexity, and communication between students and technicians.

¹ Olthaus, D. (2024, December 4). Daniel Olthaus Director of Maintenance (I. Bartels, Interviewer) [Review of Daniel Olthaus Director of Maintenance].

1. Solution Design #1

For our first solution, we propose using an app to automate the process of receiving requests, along with creating a feature for real-time updates. These real-time updates will help students better prepare for the arrival of a technician, along with providing an opportunity for technicians to communicate with students to collect additional information. It is an easy app to use, with only a few buttons for technicians to press to create messages and a direct messaging feature if needed to complete an unclear request. Our application, MU Maintenance, uses information systems to decrease complexity and increase transparency within the process of scheduling and fulfilling a maintenance request.

To better visualize what the app might look like, we created a mockup design using Photoshop and Adobe Illustrator, crafting a detailed representation of the user interface and overall functionality. The app's design was based heavily on the Business Process Model and Notation diagrams we developed for the To-Be aspect. These diagrams provided a structured, step-by-step outline of the processes the app needed to support.

Mockups, in this context, served as a bridge between the theoretical processes outlined in the BPMN and the tangible experience for the technician we aimed to create. The BPMN acted as a blueprint, guiding the layout of screens, navigation paths, and the placement of interactive elements. This methodical approach not only enhanced the feasibility of the app but also allowed us to anticipate how technicians might interact with it in real-world scenarios. By aligning the mockup with the BPMN, we could identify potential pain points or redundancies early, refining the design to better streamline the intended workflows. This process ultimately resulted in

a mockup that not only looked polished but also effectively embodied the app's core purpose, making it a helpful tool for visualizing our solution.

To begin the process, a student will submit a maintenance request and will receive an automated text message, rather than an email, to confirm that the request was received by the maintenance team. Next, the request will be sent to a scheduling team who will assign the job to a technician based on availability, convenience, and skill levels. Once the schedule is complete, the scheduling team will send it to the technician, where it will be available to view through the app MU Maintenance. Once this is locked in by both groups of employees, the student will be notified via text of a day and time frame that the technician plans to arrive to fix the feature. On the scheduled day, the maintenance employee will press a button on the app to notify the student that they are on the way to fix their item. This allows the student a few moments to prepare if they are in the room, or they can choose to ignore the message if they are out of the room. Once the request is fulfilled, the student will receive another text notification that the item has been fixed. This comes from another press of a button on the MU Maintenance App, and the maintenance employee will move on to the next project in their queue after notifying the next student that they are on the way. We then created a new model in LucidChart to depict a condensed and streamlined version of the process. This BPMN only focuses on a smaller portion of the process which is the communication between student and technician. The link for that process can be found [here](#).

2. Solution Design #2

Our second proposed solution focuses on addressing the ambiguous nature of the current maintenance scheduling process. By improving the current maintenance portal to provide more detailed emails and enhanced request forms, we aim to reduce the guesswork students face and ensure a smoother experience.

For starters, this solution would provide clear and specific two-hour time frames to eliminate the uncertainty of broad windows of arrival. This will foster a better relationship between students and the maintenance team, as there will be fewer unexpected surprises. Technicians will still have the same responsibilities, but they will be held to a tighter schedule to improve the overall process and satisfaction of everyone involved. This solution would allow busy college students the option to plan their schedules around the maintenance visit, enabling them to be present when the maintenance staff arrives, if desired.

Cost/Benefit Analysis:

Tangible costs for our proposed solution encompass all expenses related to the development, implementation, and ongoing operation of the MU Maintenance app. These costs can be categorized into two main types: one-time initial expenses and recurring expenses. One-time/initial expenses include the costs of designing, developing, and testing the app to ensure it meets the university's needs and functions effectively. Recurring costs involve ongoing app maintenance, regular updates to improve functionality, debugging to address any issues, and potential server hosting fees to support

the app's operation. According to Business of Apps, and our analysis of the app's complexity, we categorized the MU Maintenance app into the medium complexity tier and estimate that the total tangible costs will range between \$50,000 and \$100,000 covering both the initial and recurring expenses.²

Intangible costs include the opportunity costs associated with the time and resources dedicated to app development, which could have been invested in other university projects. Additionally, time and effort will be required to train university technicians to ensure they can adequately understand and use the app effectively. Beyond these concerns, there is also the potential for dissatisfaction among members of the maintenance team who may view the app as a complication or threat to their established workflows, possibly resulting in a decline in team morale. Addressing these potential challenges will be crucial to ensuring the successful adoption and long-term integration of the app into the university's maintenance process.

Along with the implementation of the MU Maintenance app will come many benefits, including increased and simplified communication, improved efficiency, and data optimization. The main goal of this project is to streamline the touchpoint between university technicians and students, which will ultimately better prepare both parties. For example, students will have the ability to send technicians visuals of maintenance issues, and sms messages about any additional information that may be needed. On the other hand, students will receive real-time updates through sms about the technician's arrival and the maintenance task's completion. The process as a whole will be much more efficient for both parties, as students will be notified

2 Dogtiev, A. (2024, September 23). App Development Cost. Business of Apps.
<https://www.businessofapps.com/app-developers/research/app-development-cost/>

when the technician arrives/ leaves and the technicians can more efficiently plan their day based on the locations of different maintenance issues, the category of the maintenance issue, and the tools needed to fix the problem. There also will be a substantial decrease in emails between students and the maintenance scheduling team, which are often slow and redundant.

Following the implementation and steady use of the MU Maintenance app, the university will have the ability to build a comprehensive database of maintenance issues that have occurred over time. This will enable data optimization by allowing the maintenance team to analyze past issues, identify recurring patterns and trends, and strategically prepare for similar problems in the future. By leveraging this historical data, the team can improve decision-making, allocate resources more efficiently, and potentially implement measures to reduce the frequency of common issues.

Additionally, the development and general maintenance of the app do not require any raw material costs, aligning with Miami's overall environmental and sustainability goals for the future. The app will only require minimal energy usage for updates and general maintenance. In this way, the app not only enhances communication and response times but also transforms how maintenance operations are planned and executed.

While the app poses a lot of potential benefits to students, maintenance workers, and the university, there are a few potential drawbacks to switching to a new system. Firstly, maintenance employees may not want to abandon the old system that they are already familiar with in order to learn and adjust to an entirely new system. The workers also may not want to make the extra effort to alert students exactly when they are coming, as they have never done that in the past and do not want to break their old habits. Both of these drawbacks come with mitigants to reduce the concerns of maintenance

employees and ease their transition to the new app. The new app will be extremely user-friendly and easy to learn, allowing the workers to pick up the new app and adjust quickly after minimal training. Additionally, the app will be so easy to use, as the employees will only need to press one button to alert the students when they are coming, allowing them to maintain their typical workday schedule with one small addition in between room fixes. Lastly, giving students a heads up so they more accurately know when a maintenance employee is coming reduces the possibility of awkward encounters with students who are not expecting someone to come in, whether they are taking an exam, changing, or not prepared for someone to enter the room.

Conclusion:

Our BPM project aimed to address the inefficiencies and frustrations experienced by Miami University students and maintenance staff during the maintenance request process. By analyzing the current system and utilizing IT solutions, we developed two innovative approaches to enhance transparency and streamline communication throughout the process. The MU Maintenance app introduces real-time updates and direct communication, simplifying the experience for both students and maintenance employees. Additionally, our enhanced maintenance portal provides improvements by refining scheduling accuracy and communication throughout the entire process. While these solutions may face initial challenges in terms of costs and employee adaptation, their benefits far outweigh the drawbacks. Both solutions improve the student and employee experience, creating a more reliable and user-friendly maintenance process that aligns with Miami University's ambitions for innovation and student satisfaction.